

$$\begin{aligned}
& -\frac{1}{16\pi^2} \left(\frac{2}{9} C_{\phi 1\phi 2} g_2^4 LF_{3,0}[m_2] + \frac{1}{3} C_{\phi 1\phi 2} g_2^4 LF_{4,-1}[m_2] + \right. \\
& \frac{16}{45} C_{\phi 1\phi 2} g_2^4 LF_{5,-2}[m_2] + \frac{1}{36} C_{\phi 1\phi 2} g_2^2 (-9 \overline{y_e^{pr}} y_e^{pr} + 4 \sum_p g_2^2) LF_{3,0}[m_l^P] + \\
& \frac{1}{24} C_{\phi 1\phi 2} g_2^2 (6 \overline{y_e^{pr}} y_e^{pr} - 5 \sum_p g_2^2) LF_{4,-1}[m_l^{P1}] + \frac{4}{45} \sum_p C_{\phi 1\phi 2} g_2^4 LF_{5,-2}[m_l^P] + \\
& \frac{1}{12} C_{\phi 1\phi 2} g_2^2 (-9 \overline{y_d^{pr}} y_d^{pr} + 9 \overline{y_u^{pr}} y_u^{pr} + 4 \sum_p g_2^2) LF_{3,0}[m_q^P] - \\
& \frac{1}{8} C_{\phi 1\phi 2} g_2^2 (-6 \overline{y_d^{pr}} y_d^{pr} + 6 \overline{y_u^{pr}} y_u^{pr} + 5 \sum_p g_2^2) LF_{4,-1}[m_q^P] + \frac{4}{15} \sum_p C_{\phi 1\phi 2} g_2^4 LF_{5,-2}[m_q^P] + \\
& \frac{1}{9} C_{\phi 1\phi 2} g_2^4 LF_{3,0}[\tilde{\mu}] + \frac{1}{6} C_{\phi 1\phi 2} g_2^4 LF_{4,-1}[\tilde{\mu}] - \frac{8}{45} C_{\phi 1\phi 2} g_2^4 LF_{5,-2}[\tilde{\mu}] - m_1 \tilde{\mu} g_1^4 LF_{2,2,-1}[m_1, \tilde{\mu}] - \\
& \frac{4}{3} C_{\phi 1\phi 2} g_2^4 LF_{2,1,0}[m_2, \tilde{\mu}] - \frac{1}{12} g_2^4 (13 C_{\phi 1\phi 2} + 36 m_2 \tilde{\mu}) LF_{2,2,-1}[m_2, \tilde{\mu}] + \\
& \frac{1}{6} m_2 g_2^4 (-3 m_2 C_{\phi 1\phi 2} + 2 \tilde{\mu} (3 C_{\phi 1^2} + C_{\phi 2^2})) LF_{2,2,0}[m_2, \tilde{\mu}] + \\
& \frac{2}{3} C_{\phi 1\phi 2} g_2^4 LF_{3,1,-1}[m_2, \tilde{\mu}] + \frac{2}{3} C_{\phi 1\phi 2} g_2^4 LF_{3,2,-2}[m_2, \tilde{\mu}] - \\
& \frac{1}{6} m_2 \tilde{\mu} g_2^4 (3 C_{\phi 1^2} + C_{\phi 2^2}) LF_{3,2,-1}[m_2, \tilde{\mu}] - \frac{1}{2} \tilde{\mu} g_1^2 \overline{y_d^{pr}} a_d^{pr} LF_{2,1,0}[m_d^r, m_q^P] - \\
& \frac{1}{4} g_1^2 (C_{\phi 1\phi 2} \overline{a_d^{pr}} a_d^{pr} + \tilde{\mu} \overline{y_d^{pr}} (a_d^{pr} (C_{\phi 1^2} + C_{\phi 2^2}) + C_{\phi 1\phi 2} \tilde{\mu} y_d^{pr})) LF_{2,2,0}[m_d^r, m_q^P] - \\
& \frac{1}{2} \tilde{\mu} g_1^2 \overline{y_e^{pr}} a_e^{pr} LF_{2,1,0}[m_e^r, m_l^P] - \\
& \frac{1}{4} g_1^2 (C_{\phi 1\phi 2} \overline{a_e^{pr}} a_e^{pr} + \tilde{\mu} \overline{y_e^{pr}} (a_e^{pr} (C_{\phi 1^2} + C_{\phi 2^2}) + C_{\phi 1\phi 2} \tilde{\mu} y_e^{pr})) LF_{2,2,0}[m_e^r, m_l^P] + \\
& \frac{1}{2} \tilde{\mu} g_1^2 \overline{y_e^{pr}} a_e^{pr} LF_{2,1,0}[m_l^P, m_e^r] - \frac{1}{4} \tilde{\mu} \overline{y_e^{pr}} a_e^{pr} (g_1^2 + g_2^2) LF_{3,1,-1}[m_l^P, m_e^r] + \\
& \frac{1}{2} (C_{\phi 1\phi 2} \overline{a_e^{pr}} a_e^{pr} (g_1^2 - g_2^2) + 2 \tilde{\mu} g_1^2 \overline{y_e^{pr}} (a_e^{pr} (C_{\phi 1^2} + C_{\phi 2^2}) + C_{\phi 1\phi 2} \tilde{\mu} y_e^{pr})) LF_{3,1,0}[m_l^P, m_e^r] + \\
& \frac{1}{4} g_1^2 (C_{\phi 1\phi 2} \overline{a_e^{pr}} a_e^{pr} + \tilde{\mu} \overline{y_e^{pr}} (a_e^{pr} (C_{\phi 1^2} + C_{\phi 2^2}) + C_{\phi 1\phi 2} \tilde{\mu} y_e^{pr})) LF_{3,2,-1}[m_l^P, m_e^r] + \\
& \frac{1}{8} (3 C_{\phi 1\phi 2} \overline{a_e^{pr}} a_e^{pr} (-g_1^2 + g_2^2) - \\
& \quad \tilde{\mu} \overline{y_e^{pr}} (3 a_e^{pr} (C_{\phi 1^2} + C_{\phi 2^2}) (3 g_1^2 + g_2^2) + C_{\phi 1\phi 2} \tilde{\mu} y_e^{pr} (9 g_1^2 + g_2^2))) LF_{4,1,-1}[m_l^P, m_e^r] + \\
& \frac{1}{6} (-C_{\phi 1\phi 2} g_2^2 \overline{a_e^{pr}} a_e^{pr} + \tilde{\mu} \overline{y_e^{pr}} (a_e^{pr} (3 g_1^2 (C_{\phi 1^2} + C_{\phi 2^2}) + 2 C_{\phi 2^2} g_2^2) + 3 C_{\phi 1\phi 2} \tilde{\mu} g_1^2 y_e^{pr})) \\
& \quad LF_{5,1,-2}[m_l^P, m_e^r] + \frac{1}{2} \tilde{\mu} g_1^2 \overline{y_d^{pr}} a_d^{pr} LF_{2,1,0}[m_q^P, m_d^r] - \\
& \frac{3}{4} \tilde{\mu} \overline{y_d^{pr}} a_d^{pr} (g_1^2 + g_2^2) LF_{3,1,-1}[m_q^P, m_d^r] + \\
& \frac{1}{4} (-C_{\phi 1\phi 2} \overline{a_d^{pr}} a_d^{pr} (g_1^2 + 3 g_2^2) + 2 \tilde{\mu} g_1^2 \overline{y_d^{pr}} (a_d^{pr} (C_{\phi 1^2} + C_{\phi 2^2}) + C_{\phi 1\phi 2} \tilde{\mu} y_d^{pr})) \\
& \quad LF_{3,1,0}[m_q^P, m_d^r] + \frac{1}{4} g_1^2 (C_{\phi 1\phi 2} \overline{a_d^{pr}} a_d^{pr} + \tilde{\mu} \overline{y_d^{pr}} (a_d^{pr} (C_{\phi 1^2} + C_{\phi 2^2}) + C_{\phi 1\phi 2} \tilde{\mu} y_d^{pr})) \\
& \quad LF_{3,2,-1}[m_q^P, m_d^r] + \frac{3}{8} (C_{\phi 1\phi 2} \overline{a_d^{pr}} a_d^{pr} (g_1^2 + 3 g_2^2) - \\
& \quad \tilde{\mu} \overline{y_d^{pr}} (a_d^{pr} (C_{\phi 1^2} + C_{\phi 2^2}) (5 g_1^2 + 3 g_2^2) + C_{\phi 1\phi 2} \tilde{\mu} y_d^{pr} (5 g_1^2 + g_2^2))) LF_{4,1,-1}[m_q^P, m_d^r] + \\
& \frac{1}{2} (-C_{\phi 1\phi 2} g_2^2 \overline{a_d^{pr}} a_d^{pr} + \tilde{\mu} \overline{y_d^{pr}} (a_d^{pr} (3 g_1^2 (C_{\phi 1^2} + C_{\phi 2^2}) + 2 C_{\phi 2^2} g_2^2) + 3 C_{\phi 1\phi 2} \tilde{\mu} g_1^2 y_d^{pr})) \\
& \quad LF_{5,1,-2}[m_q^P, m_d^r] + \frac{3}{4} C_{\phi 1\phi 2} \overline{y_u^{st}} y_u^{pt} (\overline{y_d^{pr}} y_d^{sr} - 2 \overline{y_u^{pr}} y_u^{sr}) LF_{2,1,0}[m_q^P, m_q^S] - \\
& \frac{3}{4} C_{\phi 1\phi 2} \overline{y_u^{st}} y_u^{pt} (\overline{y_d^{pr}} y_d^{sr} - 2 \overline{y_u^{pr}} y_u^{sr}) LF_{3,1,-1}[m_q^P, m_q^S] - \\
& \frac{1}{4} \tilde{\mu} \overline{y_u^{pr}} a_u^{pr} (g_1^2 - 3 g_2^2) LF_{2,1,0}[m_q^P, m_u^r] + \frac{1}{8} (-C_{\phi 1\phi 2} \overline{a_u^{pr}} a_u^{pr} (g_1^2 - 9 g_2^2) + \tilde{\mu} \overline{y_u^{pr}} \\
& \quad (a_u^{pr} (-g_1^2 (C_{\phi 1^2} + C_{\phi 2^2}) + g_2^2 (9 C_{\phi 1^2} + 5 C_{\phi 2^2})) - C_{\phi 1\phi 2} \tilde{\mu} y_u^{pr} (g_1^2 - 5 g_2^2))) LF_{2,2,0}[m_q^P, \\
& \quad m_u^r] + \frac{3}{4} g_2^2 (C_{\phi 1\phi 2} \overline{a_u^{pr}} a_u^{pr} + \tilde{\mu} \overline{y_u^{pr}} (a_u^{pr} (3 C_{\phi 1^2} + C_{\phi 2^2}) + C_{\phi 1\phi 2} \tilde{\mu} y_u^{pr})) LF_{3,1,0}[m_q^P, m_u^r] - \\
& \frac{1}{4} g_2^2 (3 C_{\phi 1\phi 2} \overline{a_u^{pr}} a_u^{pr} + 2 \tilde{\mu} \overline{y_u^{pr}} (a_u^{pr} (3 C_{\phi 1^2} + C_{\phi 2^2}) + C_{\phi 1\phi 2} \tilde{\mu} y_u^{pr})) LF_{3,2,-1}[m_q^P, m_u^r] - \\
& \frac{3}{4} g_2^2 (C_{\phi 1\phi 2} \overline{a_u^{pr}} a_u^{pr} + \tilde{\mu} \overline{y_u^{pr}} (a_u^{pr} (3 C_{\phi 1^2} + C_{\phi 2^2}) + C_{\phi 1\phi 2} \tilde{\mu} y_u^{pr})) LF_{4,1,-1}[m_q^P, m_u^r] + \\
& \frac{3}{4} C_{\phi 1\phi 2} \overline{y_d^{sr}} y_d^{st} \overline{y_u^{st}} y_u^{pt} LF_{2,1,0}[m_q^S, m_q^P] - \frac{3}{4} C_{\phi 1\phi 2} \overline{y_d^{sr}} y_d^{st} \overline{y_u^{st}} y_u^{pt} LF_{3,1,-1}[m_q^S, m_q^P] + \\
& \frac{1}{4} \tilde{\mu} \overline{y_u^{pr}} a_u^{pr} (7 g_1^2 + 3 g_2^2) LF_{2,1,0}[m_u^r, m_q^P] - \frac{3}{4} \tilde{\mu} \overline{y_u^{pr}} a_u^{pr} (g_1^2 + g_2^2) LF_{3,1,-1}[m_u^r, m_q^P] + \frac{1}{4} \\
& \quad (7 C_{\phi 1\phi 2} g_1^2 \overline{a_u^{pr}} a_u^{pr} + \tilde{\mu} \overline{y_u^{pr}} (a_u^{pr} (7 g_1^2 (C_{\phi 1^2} + C_{\phi 2^2}) + 2 C_{\phi 2^2} g_2^2) + C_{\phi 1\phi 2} \tilde{\mu} y_u^{pr} (4 g_1^2 - g_2^2))) \\
& \quad LF_{3,1,0}[m_u^r, m_q^P] + \frac{1}{8} (C_{\phi 1\phi 2} \$$